

Guidelines for MTech Project Proposal Writing

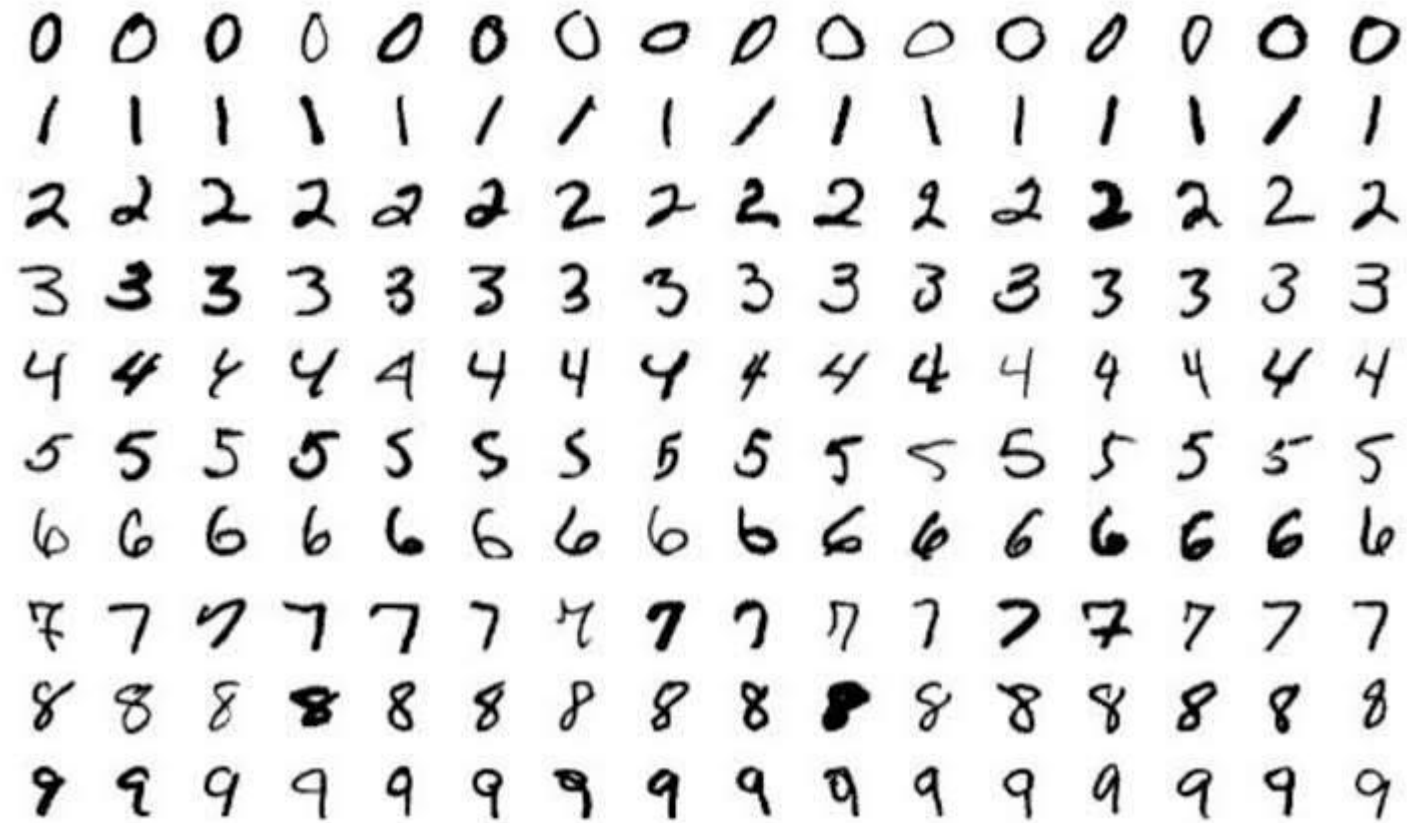
Speaker: Tirthankar Sengupta

Date: 17th July 2026



Web Enabled MTech in Industrial AI at IIT Madras

The Topic: Image classification



Overall Goal

Highlight the contribution of your work

MTech Proposal Structure

❑ Project Summary

- Introduce the topic: What is image classification using deep learning
- Use case: Scanning postal ZIP codes on envelopes
- Motivation: Why is identification of postal ZIP codes an important problem
- What are the challenges in the area ? ROI detection, Noise removal, Segmentation, etc.
- State of the art in in Optical Character Recognition (OCR) using deep learning
- Contribution of the work: Algorithm for image classification
- Specifics of contribution: Novelty, Computational cost, Edge device deployment, etc.

❑ Objectives

- Develop an Optical Character Recognition (OCR) algorithm
- Develop a computationally efficient algorithm deployable on an edge device
- Algorithm performance metrics at par with current state of the art algorithms

MTech Proposal Structure

☐ **Current State of The Art**

- Latest in Optical Character Recognition (OCR) using deep learning

☐ **Work Plan**

- Details of the plan of action for achieving the project objectives.
- Modularity: Divide the entire work plan into work packages.

☐ **Data Sets**

- List the datasets
- Details of each dataset (source, public/private, etc.)
- Details of data generation (if applicable)

MTech Proposal Structure

☐ Expected deliverables/outcomes

- OCR algorithm
- Computationally efficient model deployable on an edge device
- Comparative performance evaluation report against current state of the art models

☐ Significance of the Expected Outcome(s) with respect to the State-of-the-Art

- OCR algorithm contributions: ROI detection, Noise removal, Segmentation, etc.
- Lower computational cost when compared to current state-of-the-art models
- Superior performance when compared to current state-of-the-art model

MTech Proposal Structure

- ☐ **Implementation arrangements proposed for the project (linkages and management structure)**
 - Plan of project execution (collecting data, obtaining required resources for implementing the project, progress review mechanism, etc.)
- ☐ **Resource Requirements and their availability**
- ☐ **Risks**
- ☐ **Gantt chart**

Thank You